

Dynamic ISF — can sensitivity be derived from data? (feasibility findings)

2026-06-07 · Tim Street / Claude · companion to the methodology and v1/v2 analysis

Question

Can a user's insulin sensitivity (ISF) be derived from device-observable data well enough to remove the need to enter it — and, if so, how? This summarises a six-phase feasibility study on 138 open-source AID users (Trio, AAPS, OpenAPS), reusing the empirical-ISF ΔIOB estimator and the $\sqrt{\text{TDD}}$ law established in the equation-fit analysis.

The two separable parts of ISF

ISF has a shape, meaning how it scales with TDD and glucose, and a per-user level. The two behave very differently.

The shape is derivable and settled. Sensitivity scales with total daily dose as roughly $1/\sqrt{\text{TDD}}$, and that holds across users, with the glucose curve a separate dimension on top. The shape needs no input from anyone.

The level is the hard part, and the subject of this study. A single global level leaves the typical person about 57% off, so the question is whether the per-user level can be recovered from data rather than entered.

The feasibility arc (Phases 1–6)

phase	question	finding
1	Is a short-window empirical ISF dosing-grade?	No. $\pm 10\%$ CI only at 60–90 d; a 14-day estimate swings $\pm 34\%$ block-to-block, memoryless (lag-1 ACF ≈ 0).
2	Is the variation trackable structure?	Mostly not. Circadian $\sim 2\text{--}4\%$, weekly $\sim 1\%$, slow $\sim 2\%$; $\sim 90\%$ residual, near-white , partly measurement noise.

phase	question	finding
3	Can a stable baseline be recovered by shrinkage?	Yes, in self-consistency terms: cold-start $\pm 42\%$; shrinking own-data toward the $\sqrt{\text{TDD}}$ prior $\rightarrow \pm 22\%$ (14 d) $\rightarrow \pm 13\%$ (90 d).
5	Does the derived (sensitivity-anchored) level beat tuned ISF on real outcomes?	No — and it's unsafe. Users dosing far “weaker” than their measured sensitivity have more hypoglycaemia, not less (logR-vs-TBR +0.38). The empirical ISF is hypo-biased.
6	Does overnight data de-bias the sensitivity estimate?	No. Overnight is noisier ($\pm 50\%$ test-retest) and carries the same bias (+0.39). The bias is intrinsic to observational sensitivity, not a daytime artefact.
7	Can outcome-based tuning give a safe per-user level?	Yes, in principle. Working-anchored cold start ($355/\sqrt{\text{TDD}}$) lands $\pm 28\%$ of the users' tuned ISF with no entry; a bounded outcome nudge removes the residual hypo-signature (logR-vs-TBR $+0.21 \rightarrow -0.04$) where the sensitivity route made it worse (+0.38).

What this settles

1. The $\sqrt{\text{TDD}}$ shape is derivable and needs no entry. Anchored to working (tuned) profiles the constant is $K \approx 355$ ($\text{ISF} \approx 355/\sqrt{\text{TDD}}$); anchored to measured sensitivity it is ≈ 145 . Since Phases 5 and 6 show the measured anchor is unsafe, the working-anchored 355 is the right cold start.
2. A zero-entry cold start is feasible and reasonably good. From observable TDD alone, $\text{ISF} \approx 355/\sqrt{\text{TDD}}$ places a brand-new user within $\pm 28\%$ of the value they would have tuned to (Phase 7). Against the measured empirical baseline the figure is $\pm 42\%$, but that baseline is the hypo-biased one, so the working-ISF comparison is the number that matters for deployment.
3. The per-user level cannot be safely derived from observed sensitivity, in any window. The ΔIOB sensitivity estimate is entangled with the user's existing insulin excess or deficit: it reads as “very sensitive” (low ISF) precisely for people who run low, so dosing to it would give the most

insulin to the most hypo-prone. Restricting to clean overnight data does not fix this. It is intrinsic to measuring sensitivity from observational glucose against insulin.

4. Beyond a stable baseline there is little left to track for correction ISF (Phase 2). The right architecture is therefore not an elaborate adaptive ISF but a stable baseline plus the $\sqrt{\text{TDD}}$ demand-track, with autosens covering the bounded sub-daily and safety role it already serves.

The constructive conclusion

A data-derived, zero-entry ISF is feasible **only in this form**:

$\sqrt{\text{TDD}}$ shape (universal) + working-anchored cold start ($\approx 355/\sqrt{\text{TDD}}$, $\pm 42\%$, no entry) + bounded outcome-based self-tuning of the level — nudging ISF weaker when unexplained lows appear and stronger when sustained unexplained highs appear, damped and clamped.

It is **not** feasible as “measure the user’s sensitivity and dose to it” — that route is hypo-biased regardless of how clean the window is. The level must be tuned from **outcomes** (the autotune/autosens philosophy), not from a sensitivity regression. That is the autotune design, now empirically explained: it is built on the human-tuned anchor and outcome-attribution precisely because the measurement route fails.

Caveats

Decision-level and observational (no closed-loop simulation); single cohort, mostly users who never ran dynamic ISF; the empirical estimator carries carb/endogenous-glucose confounds; “level” errors in Phase 3 are self-consistency, not outcome-validated correctness. A closed-loop trial is required before any of this drives dosing. Nothing here is medical advice.

Reproducibility

`inv008/phase{1,2,3,5,6}*.py` and `results/phase*_{*.json,md}` in `github.com/tim2000s/dynamic-isf-calculations`. Inputs device-observable; the user’s profile ISF is used as a benchmark only.

Addendum, 27 August 2026: replication on independent data

Phases 5 and 6 above concluded that observational sensitivity is biased low, that dosing to it is unsafe, and that the bias is intrinsic rather than a daytime artefact. Those conclusions rested on 138 users of open-source systems with insulin on board taken from the loop's own record.

They have since been tested on 1,679 people across six Jaeb trial cohorts, ages 2 to 82, with insulin on board reconstructed from delivery records rather than reported. The result does not carry over unchanged. Running the construction used here, the fall over the horizon divided by insulin on board at the start, gives a ratio of measured to entered sensitivity of 0.72 to 1.12 on those cohorts, and the open-loop cohort exceeds its entered settings at six hours. Constructions that instead divide by the action of all insulin above the programmed basal give 0.30 to 0.66 on the same records.

The 0.41 reported here does not survive re-examination on its own records either. Its two anchors were fitted on different subsets, 114 people for the measured constant and 138 for the entered one, and both were regression fits rather than medians. Matched person by person across the same 112 records and taken as medians, the constants are 207 and 333, a ratio of 0.62, and the median of per-person ratios is 0.60. As a Walsh-style constant that is 1432 measured against 2219 entered.

So the ratio of 0.41 reported here is not reproduced at that value on trial data, and the spread between constructions on identical records is wider than the spread between cohorts. Why theoref archive sits at 0.41 where trial data sits near 1.0 under the same construction is open, and is a question about these records rather than about the method. The conclusions of Phases 5 and 6, that the measured level is unsafe to dose to and that overnight does not de-bias it, are untouched by this: they rest on outcomes rather than on the ratio. See Dynamic-ISF-7.